



USE CASE 5 – EMAN CHAUDHARY

Industrial Predictive Maintenance System

Domain: Oil & Gas / Energy

Assessment Type	AI/ML Engineering Take-Home Assignment
Time Allowed	2 weeks from date of issue
Presentation	Online interview, candidate presents, followed by panel Q&A
Prototype	Optional, earns bonus points if demonstrated live or via recording

1. Background & Context

An energy processing company operates critical rotating equipment — pumps, compressors, gas turbines, and centrifuges, at multiple upstream and midstream facilities. Unplanned equipment failure causes production halts costing \$50,000 to \$100,000 per hour, in addition to significant safety and environmental risk. Current maintenance is time-based: equipment is serviced on a fixed calendar schedule regardless of its actual condition.

The company wants to transition to condition-based predictive maintenance, using real-time sensor data to estimate equipment health and predict failures before they occur. This would allow maintenance to be scheduled at the optimal moment, extending equipment life while preventing costly unplanned breakdowns.

2. Problem Statement

- Ingest and process multi-variate sensor time-series data (vibration, temperature, oil pressure, RPM) from industrial equipment assets.
- Detect early signs of equipment degradation and predict the probability of failure within a defined time window (e.g., next 24 or 72 hours).
- Estimate Remaining Useful Life (RUL) for each monitored asset.
- Generate automated maintenance alerts with recommended actions when risk crosses defined thresholds.

- Produce a natural language maintenance briefing report per asset for non-technical operations managers.

3. Your Assignment

Design, and optionally prototype, an Industrial Predictive Maintenance System. Your solution should include all four components below.

Component A: Data Ingestion & Feature Engineering

- Ingest time-series sensor readings for multiple equipment units across at least two failure modes.
- Engineer health indicators from raw signals: rolling statistics, rate-of-change metrics, and (for vibration data) FFT-based frequency features if feasible.
- Define and justify your target variable: binary failure prediction within a time window, or continuous RUL estimation.

Component B: Predictive Models

- Train models for (a) anomaly/degradation detection and (b) failure probability prediction or RUL estimation.
- Compare at least two modelling approaches and select the best based on appropriate evaluation metrics.
- Track all experiments, parameters, and metrics using MLflow.
- Discuss the trade-off between model interpretability and predictive performance for this safety-critical use case.

Component C: Alert & Escalation System

- Define risk thresholds and design how maintenance alerts are triggered, classified, and escalated.
- Each alert should include: Asset ID, current sensor readings, predicted risk score, time to predicted failure, and recommended maintenance action.
- Consider how the alert system would integrate with a CMMS (Computerised Maintenance Management System) in a real deployment.

Component D: Dashboard & AI Maintenance Briefing

- A Streamlit dashboard showing equipment health scores per asset, trend charts, active alerts, and RUL estimates.
- Use an LLM to generate a short, plain-language maintenance briefing per asset. Example: "Pump P-112 shows elevated bearing vibration. Predicted failure probability within 72 hours: 34%. Recommend inspection of bearing assembly and lubrication check before next shift."

4. Suggested Data Sources

- NASA CMAPSS Turbofan Engine Degradation Simulation Dataset: the classic RUL benchmark, free from NASA and UCI Machine Learning Repository.
- AI4I 2020 Predictive Maintenance Dataset (UCI): multi-failure-mode dataset with 10,000 records. Free download.
- MIMII Dataset (Zenodo): sounds of industrial machines for anomaly detection using audio signals. Free download.
- PHM Society Data Challenge datasets: various equipment failure datasets, many freely available.

5. Suggested Technology Stack

Component	Suggested Options (Open-Source / Free Tier)
Data Processing	Python, pandas, NumPy, SciPy (signal processing and FFT features)
ML / Deep Learning	scikit-learn (Random Forest, SVM, Gradient Boosting); PyTorch or TensorFlow (LSTM)
RUL Estimation	Survival analysis via the lifelines library (free); regression-based RUL models
Experiment Tracking	MLflow (free, open-source)
Dashboard	Streamlit (free); optional Grafana + InfluxDB for time-series visualisation (both free)
LLM for Briefings	Ollama with Llama 3 or Mistral (local, completely free)
Bonus: Enterprise Platform	Databricks Community Edition, IBM watsonx.ai free tier

6. Deliverables

Your submission is a structured presentation delivered in the online interview. The table below lists what the Kermit Tech panel expects to see:

Presentation Section	What We Are Looking For
1. Problem Overview	Your understanding of the business context and the core challenge
2. Assumptions	What you assumed about data, users, environment, scale, and constraints

3. Proposed Solution	Your chosen approach, the components you selected, and why
4. Architecture Diagram	A clear end-to-end diagram showing data flow, components, and integrations
5. Data Engineering	How data is ingested, transformed, stored, and served
6. AI/ML Methodology	Model selection rationale, training approach, and evaluation metrics
7. GenAI / RAG Component	How GenAI is used, prompt strategy, retrieval design, and output quality
8. Security Considerations	Data security, access control, API security, and PII handling
9. Scalability & Robustness	How the solution handles higher data volumes, failures, and edge cases
10. Limitations & Next Steps	Honest assessment of what is missing and how you would improve it
Prototype (Optional Bonus)	Live demo or recorded walkthrough of a working implementation

7. Special Instructions & Conditions

- All platforms and tools must be free tier, open-source, or community/trial edition. You should not incur any personal cost to complete this assignment.
- You are welcome to use commercial platform free trials (Snowflake 30-day trial, Databricks Community Edition, IBM watsonx.ai free tier, Denodo Express) as bonus components, but your core solution must function without them.
- Architecture diagrams may be created with any free tool: draw.io / diagrams.net, Mermaid, Excalidraw, Lucidchart free tier, or hand-drawn and photographed clearly.
- For the LLM / GenAI component, use Ollama (local, completely free) with Llama 3 or Mistral, or the HuggingFace free inference API. Do not use paid OpenAI or Anthropic API tiers.
- You are encouraged to generate synthetic data where real datasets are insufficient. Document clearly how you generated it, including the parameters and distributions you used.
- If using public datasets, cite your sources in the presentation.
- Your presentation should be no longer than 20-25 slides or equivalent. Quality and clarity over quantity.
- You will have 20 minutes to present, followed by 15-20 minutes of Q&A with the Kermit Tech panel. Be prepared to answer detailed technical questions on every component you present.

Confidential — This document is issued exclusively to the named candidate and must not be shared with others.